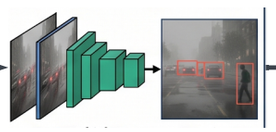
1. Feature Extraction

Input Camera Frame from CARLA-SUMO



frame, detector

On-Vehicle Feature Extraction Runs backbone and detection head of on to obtain features and detections



On-Vehicle Feature Extraction

Runs backbone and detection head of m_i on x_i to obtain features F_i and detections

2. Uncertainty Assessment

Compute Predictive Uncertainty

Calculate u_i^{pred} using Equation (9) τ from Eq. (12)

$$u_i^{pred} \leftarrow 1\ P'(x_i)$$

$$u_i^{\{pred\}} \leftarrow 1\ if P_i = \theta$$



$u_i^{pred} = 0.45$

High uncertainty value because

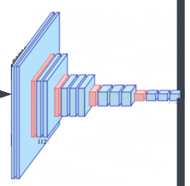
dense smog

3. Gating Logic

25% budget trigger

validation-calibrated threshold

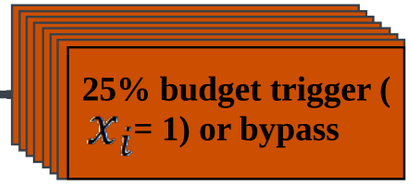




$x_i \leftarrow 1$ True

False $x_i \leftarrow 0$

$x_i \leftarrow 1$

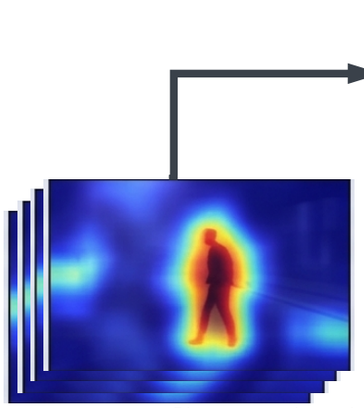


25% budget trigger (x_i = 1) or bypass

4. Gated Path Choice

Uncertain Task (x_i = 1)

(Active dual-head XAI feedback)



Heat-map/salient activation
Active evidence calibration

Compute Saliency $\hat{S}_i$ & Alignment Reliability r_i (Eq. 10)

Fused Feedback Generation Compute fused score

Certain Task (x_i= 0)

(No gating)

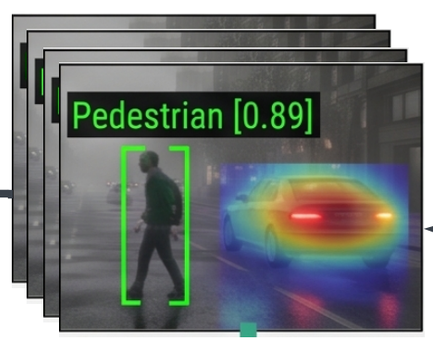
Saliency Head Bypassed

Skips the costly saliency head to save computation (15.8 ms, 95 MB GPU space) when $u_i^{pred} < \tau$

Retain Default Metrics

5. Output & Feedback

Task Output Return



Outcome Output Card



Fidelity: 0.84
Cost per active call: 15.8 ms
Amortized cost: 3.95 ms (25% budget)